Assignment 7 Solution

1. Create a directory “EmployeeDetails” on HDFS “/” using Hadoop Shell commands. (2 marks)

```
root@namenode:/# hdfs dfs -mkdir /EmployeeDetails
```

2. Copy the file “EmployeeData” from your local system to the docker container where NameNode of HDFS is started. (2 marks)

```
C:\Users\HP>docker cp "C:\Users\HP\Downloads\Module_7_Assignment_Question\Module 7 Assignment Question\EmployeeData.txt"
 namenode:/tmp/EmployeeData.txt
Successfully copied 3.58kB to namenode:/tmp/EmployeeData.txt
```

3. Load the file “EmployeeData” from your local container to HDFS directory “/EmployeeDetails” created above. (2 marks)

```
C:\Users\HP>docker exec -it namenode bash
root@namenode:/# hdfs dfs -put /tmp/EmployeeData.txt /EmployeeDetails/EmployeeData.txt
2026-04-30 06:40:08,241 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
```

4. Show the list of all files in the directory “/EmployeeDetails” (2 marks)

```
root@namenode:/# hdfs dfs -ls /EmployeeDetails
Found 1 items
-rw-r--r--   1 root supergroup       1717 2026-04-30 06:40 /EmployeeDetails/EmployeeData.txt
root@namenode:/#
```

5. Write a PySpark program to perform the following operations: (3 * 4 = 12 marks)

a) Create a DataFrame using the file “/EmployeeDetails/EmployeeData” from HDFS as stored in the previous question and display top 10 entries.

```
>>> from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DoubleType
>>>
>>> schema = StructType([
...     StructField("employeeId", IntegerType(), True),
...     StructField("employeeAge", IntegerType(), True),
...     StructField("salary", DoubleType(), True),
...     StructField("country", StringType(), True)
... ])
>>>
>>> df = spark.read.csv("hdfs://namenode:9000/EmployeeDetails/EmployeeData.txt", schema=schema, header=False)
>>> df.show(10)
+----------+-----------+--------+-------+
|employeeId|employeeAge|  salary|country|
+----------+-----------+--------+-------+
|         1|         23| 50886.0|  India|
|         2|         27| 56915.0|  India|
|         3|         23| 58335.0|  India|
|         4|         26| 80492.0|    USA|
|         5|         29| 61421.0|  India|
|         6|         22|110027.0|    USA|
|         7|         20| 40059.0|  India|
|         8|         23| 33926.0|  India|
|         9|         20|103426.0|    USA|
|        10|         22| 75736.0|    USA|
+----------+-----------+--------+-------+
only showing top 10 rows
```

b) Display the count of employees country wise using the dataframe created above.

```
>>> df.groupBy("country").count().show()
+-------+-----+
|country|count|
+-------+-----+
|  India|   49|
|    USA|   51|
+-------+-----+
```

c) Display average salary paid to employees age-wise.

```
>>> from pyspark.sql.functions import avg
>>> df.groupBy("employeeAge").agg(avg("salary").alias("avg_salary")).orderBy("employeeAge").show()
+-----------+------------------+
|employeeAge|        avg_salary|
+-----------+------------------+
|         20|           65946.2|
|         21| 84698.44444444444|
|         22| 75671.36363636363|
|         23|44700.666666666664|
|         24| 82034.76923076923|
|         25|           77459.0|
|         26| 76328.84615384616|
|         27| 70208.53846153847|
|         28|          87261.375|
|         29|           87498.7|
+-----------+------------------+
```

6. Create Hive table as per the fields of the file “EmployeeData” and load the records from HDFS. (2 marks)

```
>>> spark.sql("INSERT INTO empdb.EmployeeData VALUES (9999, 28, 72000.0, 'India')")
DataFrame[]
>>> spark.sql("SELECT * FROM empdb.EmployeeData").show()
+----------+-----------+--------+-------+
|employeeId|employeeAge|  salary|country|
+----------+-----------+--------+-------+
|         1|         23| 50886.0|  India|
|         2|         27| 56915.0|  India|
|         3|         23| 58335.0|  India|
|         4|         26| 80492.0|    USA|
|         5|         29| 61421.0|  India|
|         6|         22|110027.0|    USA|
|         7|         20| 40059.0|  India|
|         8|         23| 33926.0|  India|
|         9|         20|103426.0|    USA|
|        10|         22| 75736.0|    USA|
|        11|         21|115368.0|    USA|
|        12|         27| 76429.0|    USA|
|        13|         22| 41530.0|  India|
|        14|         22| 85123.0|    USA|
|        15|         27| 23135.0|  India|
|        16|         29| 99802.0|    USA|
|        17|         22| 43058.0|  India|
|        18|         29|118167.0|    USA|
|        19|         23| 38456.0|  India|
|        20|         21| 98042.0|    USA|
+----------+-----------+--------+-------+
only showing top 20 rows
```

7. Write a spark program to connect to Hive and display all records. Write code to insert a record to the Hive table using Spark and display. (8 marks)

```
from pyspark.sql import SparkSession

from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DoubleType

# ------------------------------------------------------

 Create SparkSession with Hive Support

# ------------------------------------------------------

spark = SparkSession.builder \

.appName("SparkHiveEmployeeApp") \

.enableHiveSupport() \

.getOrCreate()

print("=" * 55)

print(" Spark Session Created Successfully")

print("=" * 55)


# ------------------------------------------------------

Create Database and Table (if not exists)

# ------------------------------------------------------

spark.sql("CREATE DATABASE IF NOT EXISTS empdb")

spark.sql("USE empdb")

spark.sql("""

CREATE TABLE IF NOT EXISTS EmployeeData (

employeeId INT,

employeeAge INT,

salary DOUBLE,

country STRING

)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ','

STORED AS TEXTFILE

""")
```

```
# Load data from HDFS into the Hive table
# (Uses DataFrame write to avoid FS mismatch issues)
schema = StructType([
StructField("employeeId", IntegerType(), True),
StructField("employeeAge", IntegerType(), True),
StructField("salary", DoubleType(), True),
StructField("country", StringType(), True)
])

df = spark.read.csv(
"hdfs://namenode:9000/EmployeeDetails/EmployeeData.txt",
schema=schema,
header=False
)

df.write.mode("overwrite").saveAsTable("empdb.EmployeeData")
print("\nData loaded into Hive table: empdb.EmployeeData")
# ------------------------------------------------------
# Step 3: Connect to Hive and Display ALL Records
# ------------------------------------------------------
print("\n" + "=" * 55)
print(" All Records from Hive Table (empdb.EmployeeData)")
print("=" * 55)
all_records = spark.sql("SELECT * FROM empdb.EmployeeData")
all_records.show()
print(f" Total Records: {all_records.count()}")
```

```
>>> spark.sql("SELECT * FROM empdb.EmployeeData").show()
2026-05-02 12:51:59 WARN  ObjectStore:568 - Failed to get database global_temp, returning NoSuchObjectException
+----------+-----------+--------+-------+
|employeeId|employeeAge|  salary|country|
+----------+-----------+--------+-------+
|         1|         23| 50886.0|  India|
|         2|         27| 56915.0|  India|
|         3|         23| 58335.0|  India|
|         4|         26| 80492.0|    USA|
|         5|         29| 61421.0|  India|
|         6|         22|110027.0|    USA|
|         7|         20| 40059.0|  India|
|         8|         23| 33926.0|  India|
|         9|         20|103426.0|    USA|
|        10|         22| 75736.0|    USA|
|        11|         21|115368.0|    USA|
|        12|         27| 76429.0|    USA|
|        13|         22| 41530.0|  India|
|        14|         22| 85123.0|    USA|
|        15|         27| 23135.0|  India|
|        16|         29| 99802.0|    USA|
|        17|         22| 43058.0|  India|
|        18|         29|118167.0|    USA|
|        19|         23| 38456.0|  India|
|        20|         21| 98042.0|    USA|
+----------+-----------+--------+-------+
only showing top 20 rows
```

Insert a New Record into the Hive Table

```
print("\n" + "=" * 55)
print(" Inserting New Record into Hive Table...")
print("=" * 55)
spark.sql("""
INSERT INTO empdb.EmployeeData
VALUES (9999, 28, 72000.0, 'India')
""")
print(" Record Inserted: (9999, 28, 72000.0, 'India')")
```

```
>>> spark.sql("INSERT INTO empdb.EmployeeData VALUES (9999, 28, 72000.0, 'India')")
DataFrame[]
```

Display All Records AFTER Insertion

```
print("\n" + "=" * 55)
print(" All Records AFTER Insertion")
print("=" * 55)
```

```
updated_records = spark.sql("SELECT * FROM empdb.EmployeeData")

updated_records.show()

print(f" Total Records After Insert: {updated_records.count()}")
```

```
>>> spark.sql("INSERT INTO empdb.EmployeeData VALUES (9999, 28, 72000.0, 'India')")
DataFrame[]
>>> spark.sql("SELECT * FROM empdb.EmployeeData").show()
+----------+-----------+--------+-------+
|employeeId|employeeAge|  salary|country|
+----------+-----------+--------+-------+
|         1|         23| 50886.0|  India|
|         2|         27| 56915.0|  India|
|         3|         23| 58335.0|  India|
|         4|         26| 80492.0|    USA|
|         5|         29| 61421.0|  India|
|         6|         22|110027.0|    USA|
|         7|         20| 40059.0|  India|
|         8|         23| 33926.0|  India|
|         9|         20|103426.0|    USA|
|        10|         22| 75736.0|    USA|
|        11|         21|115368.0|    USA|
|        12|         27| 76429.0|    USA|
|        13|         22| 41530.0|  India|
|        14|         22| 85123.0|    USA|
|        15|         27| 23135.0|  India|
|        16|         29| 99802.0|    USA|
|        17|         22| 43058.0|  India|
|        18|         29|118167.0|    USA|
|        19|         23| 38456.0|  India|
|        20|         21| 98042.0|    USA|
+----------+-----------+--------+-------+
only showing top 20 rows
```

```
# ------------------------------------------------------

Verify the Newly Inserted Record

# ------------------------------------------------------

print("\n" + "=" * 55)

print(" Verification - Newly Inserted Record")

print("=" * 55)

spark.sql("""

SELECT * FROM empdb.EmployeeData

WHERE employeeId = 9999

""").show()

print("\nProgram Completed Successfully!")

print("=" * 55)

# Stop SparkSession

spark.stop()
```

```
>>> spark.sql("SELECT * FROM empdb.EmployeeData WHERE employeeId = 9999").show()
+----------+-----------+-------+-------+
|employeeId|employeeAge| salary|country|
+----------+-----------+-------+-------+
|      9999|         28|72000.0|  India|
+----------+-----------+-------+-------+
```